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Switching rocker arm having stamped inner arm configuration

Abstract

A switching roller finger follower (SRFF) assembly for valve actuation includes an outer arm and an inner arm pivotally coupled to the outer arm via a pivot axle and at least partially disposed within the outer arm. The inner arm is unitarily formed and has a first inner side arm and a second inner side arm having respective laterally spaced apart longitudinal bends that each define pin apertures. A spring engagement pin extends through the pin apertures, a pair of lost motion springs are supported by the pivot axle, an inner roller is rotatably coupled to the inner arm, and a pair of outer rollers are disposed on respective axles supported by the outer arm.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/EP2021/025127 filed Apr. 6, 2021, which claims the benefit of U.S. Provisional App. No. 63/005,768, filed on Apr. 6, 2020. The disclosure of the above application is incorporated herein by reference.

FIELD
(1) The present disclosure relates generally to switchable rocker arm assemblies and, more specifically, to a switching roller finger follower (SRFF) having a stamped inner arm including longitudinal bends, a stopping feature, and a compact latching design.
BACKGROUND
(2) Switching rocker arms allow for control of valve actuation by alternating between two or more states, usually involving multiple arms, such as in inner arm and outer arm. In some circumstances, these arms engage different cam lobes, such as low-lift lobes, high-lift lobes, and no-lift lobes. Switching rocker arms can be implemented as part of systems commonly referred to as variable valve timing (VVT) or variable valve actuation (VVA) to improve fuel economy, reduce emissions and improve driver comfort over a range of speeds. Mechanisms are required for switching rocker arm modes in a manner suited for operation of internal combustion engines.
(3) Several types of the VVA rocker arm assemblies include an inner rocker arm within an outer rocker arm that are biased together with torsion springs. Switching rocker arms allow for control of valve actuation by alternating between latched and unlatched states. A latch, when in a latched position causes both the inner and outer rocker arms to move as a single unit. When unlatched, the inner and outer arms are allowed to move independent of each other. In some circumstances, these arms can engage different cam lobes, such as low-lift lobes, high-lift lobes, and no-lift lobes. Mechanisms are required for switching rocker arm modes in a manner suited for operation of internal combustion engines.
(4) The background description provided herein is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this

background section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

SUMMARY

(5) In one example aspect, a switching roller finger follower (SRFF) assembly for valve actuation is provided. The SRFF assembly includes an outer arm and an inner arm pivotally coupled to the outer arm via a pivot axle and at least partially disposed within the outer arm. The inner arm is unitarily formed and has a first inner side arm and a second inner side arm having respective laterally spaced apart longitudinal bends that each define pin apertures. A spring engagement pin extends through the pin apertures, a pair of lost motion springs are supported by the pivot axle, an inner roller is rotatably coupled to the inner arm, and a pair of outer rollers are disposed on respective axles supported by the outer arm.

(6) In addition to the foregoing, the described SRFF assembly may include one or more of the following features: wherein the respective longitudinal bends are offset from each other by a predetermined distance; wherein the spring engagement pin defines crimped regions thereon; wherein each lost motion spring has a first end that engages the respective crimped regions of the spring engagement pin, the crimped regions reducing contact stress on the respective lost motion springs; wherein the first inner side arm and the second inner side arm each have a bushing aperture to receive a bushing and an inner roller axle; and wherein the inner roller is rotatably coupled to the inner arm by the bushing.

(7) In addition to the foregoing, the described SRFF assembly may include one or more of the following features: wherein the outer arm comprises a stopping feature thereon configured to stop rotation of the inner arm relative to the outer arm; wherein the stopping feature comprises material on the outer arm defining a shoulder to be contacted by the inner arm; wherein the stopping feature comprises a stop pin extending from the outer arm; a latch assembly configured to selectively latch the inner arm to the outer arm to prevent relative movement therebetween; and wherein the latch assembly comprises a latch pin slidingly received within a latch bore formed in the outer arm.

(8) In addition to the foregoing, the described SRFF assembly may include one or more of the following features: wherein the outer arm includes a hydraulic port fluidly coupled to the latch bore; wherein the hydraulic port is configured to selectively receive pressurized fluid from a hydraulic lash adjuster; wherein the latch assembly further comprises a latch cage operably associated with the latch pin; wherein the latch cage is biased away from the latch pin via a biasing member; wherein the latch pin defines a latch cage chamber configured to selectively receive at least a portion of a finger of the latch cage; and wherein the latch cage finger includes a shelf region and wherein the latch cage comprises a flat, wherein the flat is configured to engage the shelf region of the latch cage and inhibit rotation of the latch pin.

(9) In one example aspect, a switching roller finger follower (SRFF) assembly for valve actuation is provided. The SRFF assembly includes an outer arm and an inner arm pivotally coupled to the outer arm via a pivot axle and at least partially disposed within the outer arm. The inner arm is unitarily formed and includes a first inner side arm and a second inner side arm having respective longitudinal bends that each define pin apertures. A spring engagement pin extends through the pin apertures and defines crimped regions thereon. A pair of lost motion springs are supported by the pivot axle and have ends that engage the respective crimped regions of the spring engagement pin, the crimped regions reducing contact stress on the respective lost motion springs. An inner roller is coupled to the inner arm by a bushing, and a pair of outer rollers are disposed on respective axles supported by the outer arm.

(10) In addition to the foregoing, the described SRFF assembly may include one or more of the following features: wherein the respective longitudinal bends are laterally distanced and offset from each other by a predetermined distance; and a latch assembly configured to selectively latch the inner arm to the outer arm to prevent relative movement therebetween, the latch assembly

including a latch pin slidingly received within a latch bore formed in the outer arm, the latch pin defining an internal latch cage chamber and a flat, a latch cage having a finger configured to be at least partially received within the latch cage chamber, the finger including a shelf region, wherein the flat is configured to engage the shelf region of the latch cage and inhibit rotation of the latch pin, and a biasing member configured to bias the latch pin away from the latch cage.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) FIG. **1**A is a plan view of a switching roller finger follower (SRFF) constructed in accordance to one example of the present disclosure;
(2) FIG. **1**B is an exemplary camshaft lobe configuration for cooperation with the SRFF of FIG. **1**A;
(3) FIG. **2** is a perspective view of the SRFF of FIG. **1**A;
(4) FIG. **3** is a detail view of an exemplary stopping feature configured on the outer arm of the SRFF of FIG. **2**;
(5) FIG. **4** is a perspective view of an SRFF constructed in accordance to another example of the present disclosure and incorporating a stop pin according to other features;
(6) FIG. **5** is a detail view of an exemplary stop pin configured on the outer arm of the SRFF of FIG. **4**;
(7) FIG. **6** is a side view of the example inner arm of the SRFF of FIG. **1**A;
(8) FIG. **7** is a plan view of the inner arm of FIG. **6** shown with a pin extending through pin apertures;
(9) FIG. **8** is a plan view of another inner arm that may be utilized with the SRFF and shown with material removed to accommodate tooling;
(10) FIG. **9** is a sectional view of the latch assembly taken along lines **9-9** of FIG. **1**A;
(11) FIG. **10** is a sectional view of the latch assembly taken along lines **10-10** of FIG. **9** and shown with the latch pin engaged and not rotated;
(12) FIG. **11** is a sectional view of the latch assembly of FIG. **10** and shown with the latch pin rotated until a flat on the latch pin engages a shelf region of the latch cage;
(13) FIG. **12**A is front perspective view of the latch pin of the latch assembly shown in FIG. **9**;
(14) FIG. **12**B is a rear perspective view of the latch pin shown in FIG. **12**A;
(15) FIG. **13**A is a perspective view of the latch cage of the latch assembly shown in FIG. **9**; and
(16) FIG. **13**B is an end view of the latch cage shown in FIG. **13**A.
DETAILED DESCRIPTION
(17) As will become appreciated from the following discussion, the present disclosure provides a switching roller finger follower (SRFF) assembly with a stamped inner arm having improved longitudinal bends, a stopping feature, and a compact latching design. Longitudinal bends, offset relative to each other so as not to touch are incorporated on an inner arm geometry. A reaction pin used to stop the rotation of the lost motion spring has been moved at one end of the inner arm. In one configuration, the inner arm rotation is stopped using material on the outer arm. In another configuration, a pin is incorporated on the outer arm that is configured to stop inner arm rotation. A new latch pin and cage configuration is also disclosed.
(18) With initial reference to FIGS. **1**A-**3**, a SRFF assembly constructed in accordance to one example of the present disclosure is shown and generally identified at reference numeral **10**. FIGS. **4** and **5** illustrate SRFF assembly **10** constructed in accordance to another example of the present disclosure with a different stopping arrangement, as described herein in more detail. Like parts are identified with like reference numerals. In the example embodiments, the SRFF assembly **10** generally includes an inner arm **12** and an outer arm **14**. The default configuration is in the normal-

lift (latched) position where the inner arm **12** and the outer arm **14** are locked together, causing an engine valve (not shown) to open and allowing the cylinder to operate as it would in a standard valvetrain. When a latch assembly **16** is engaged (e.g., oil from an oil control valve feeds a hydraulic lash adjuster **18**, FIG. **9**, to engage latch assembly **16**), the inner arm **12** and the outer arm **14** operate together like a standard rocker arm to open the engine valve. In the secondary (unlatched) position, the inner arm **12** and the outer arm **14** can move independently to enable the desired secondary function from the SRFF assembly **10**.

(19) In the example embodiment, the inner arm **12** and the outer arm **14** are both mounted to a pivot axle **20**, which secures the inner arm **12** to the outer arm **14** while also allowing a rotational degree of freedom pivoting about the pivot axle **20** when the SRFF assembly **10** is in the unlatched state. A pair of lost motion torsion springs **22** are secured to the pivot axle **20** and are configured to bias the position of the inner arm **12** so that it always comes back to the starting position where the related lift can start. As shown in FIG. **1**A, the outer arm **14** includes a first outer side arm **30** and a second outer side arm **32**. The first and second outer side arms **30**, **32** each include an axle **36** that supports an outer roller **38** disposed outboard of each of the first and second outer side arms **30**, **32**.

(20) With continued reference to FIGS. **1**A-**5** and additional reference to FIGS. **7** and **8** further features of the inner arm **12** will be described. As shown in FIG. **1**A, the inner arm **12** is disposed between the first outer side arm **30** and the second outer side arm **32**. The inner arm **12** includes a first inner side arm **40** and a second inner side arm **42** coupled by a connecting member **44**. The first and second inner side arms **40**, **42** each include an aperture **46** configured to receive a bushing **48** and an axle **50** therethrough. An inner roller **52** is coupled to the inner arm **12**, by means of the bushing **48** and axle **50**, between the first and second inner side arms **40**, **42**.

(21) For exemplary purposes, as shown in FIG. **1**B, a cam **54** is configured for cooperation with the SRFF assembly **10**. In the illustrated example, the cam **54** includes an inner cam **56** and a pair of outer cams **58**. The inner cam **56** is configured to engage the inner roller **52** while the outer cams **58** are configured to engage the outer rollers **38**.

(22) With continued reference to FIGS. **6-8**, in the example embodiment, the geometry of the stamped inner arm **12** advantageously provides improved stiffness and packaging of the SRFF **10**, thereby reducing the overall width in the pad/roller area. Moreover, two additional folding operations are employed to create a geometry of inner arm bends **40**A and **42**A (FIG. **7**) which reduce the width of the inner arm **12** in the valve area. By reducing the width in this area, the lost motion springs **22** can be moved outboard of the outer side arms **30**, **32** (see FIG. **1**A), thus preserving the packaging and improving the performance of the lost motion springs **22**. In particular, the improved packaging allows for increased number of coils and spring diameter in the allotted space.

(23) As illustrated in FIG. **7**, in the example implementation, the first inner side arm **40** includes the first longitudinal bend **40**A, and the second inner side arm **42** includes the second longitudinal bend **42**A. As shown, the first and second longitudinal bends **40**A and **42**A are laterally spaced or offset and do not touch such that a predetermined distance **60** is defined therebetween. In some examples, the distance **60** is greater than the tooling width. In this regard, coining or other operations may not be required (FIG. **7**). In other examples, material from the first and second longitudinal bends **40**A, **42**A can be removed with coining or other operation (FIG. **8**) to ensure enough space for the tooling to crimp a spring engagement pin **62**, as described herein in more detail.

(24) With continued reference to FIGS. **2**, **4**, and **7**, additional features of the SRFF assembly **10** will be described. In the example implementation, the spring engagement pin **62** extends through pin apertures **64** defined through the bends **40**A and **42**A. As shown, the spring engagement pin **62** defines crimped regions **66**, which are configured to receive respective lost motion springs **22**. The crimped regions **66** advantageously reduce contact stress on the lost motion springs **22**. The coining or other operations are used to remove material **68** (FIG. **7**) on the inner arm **12** if there is not enough space for the tooling to crimp the spring engagement pin **62**. However, those skilled in the

art will appreciate that the crimped regions can be provided in any suitable location along the length of the spring engagement pin **62** to accommodate the lost motion spring **22** such as, for example, if the lost motion spring **22** is configured for use inside of the inner arm **12**.

(25) With particular reference to FIG. **2**, an exemplary stopping feature **70** will be described. In examples where cam lash is not required, a stopping feature **70** can be provided via the material on the outer arm **14** such as, for example a shoulder **72**. In this regard, the stopping feature **70** can stop rotation of the inner arm **12** (e.g., counterclockwise in FIG. **2**) when the rocker arm **10** is not assembled on the engine. In other examples, shown in FIGS. **4** and **5**, if cam lash is required, another stopping feature in the form of a stop pin **80** is assembled on the outer arm **14** to preclude rotation of the inner arm **12**. Such stopping features **70**, **80** allow the removal of floating axle configurations utilized in prior art systems. In one particular example, replacement of the floating axle configuration with the configuration that includes bushing **48** and axle **50** on the inner arm **12** improves stiffness of the SRFF **10** as a whole and allows for more freedom in the position of the outer rollers **38** with regards to the inner roller **52**.

(26) Turning now to FIGS. **9-13**, the latch assembly **16** will be described in more detail. In the example embodiment, the latch assembly **16** generally includes a latch pin **100**, a latch cage **102**, and a latch biasing member **104** (e.g., a spring). The latch pin **100** is slidingly received within a latch bore **106** formed in the outer arm **14** and is configured to move between a deployed latched position (FIG. **9**) and a withdrawn unlatched position (not shown). In the latched position, the latch pin **100** is extended to engage the inner arm **12** and prevent rotation relative to the outer arm **14**. In the unlatched position, the latch pin **100** is withdrawn into the latch bore **106** and no longer engages the inner arm **12** to allow rotation relative to the outer arm **14**.

(27) As shown in FIG. **9**, the latch pin **100** defines an internal latch cage chamber **108** and outer chamber **110**. The latch cage chamber **108** is configured to receive a finger **112** of the latch cage **102**, and the outer chamber **110** defines a seat **114** configured to receive one end of the latch biasing member **104**. The latch cage **102** includes a flange **116** configured to receive and seat the other end of the latch biasing member **104**, which is configured to bias the latch pin **100** away from the latch cage **102** toward the latched position. A hydraulic port **118** (FIG. **9**) is configured to receive a pressurized fluid (e.g., oil) from HLA **18** to move the latch pin **100** toward the latch cage **102** to the unlatched position, and the latch cage includes a cutout or window **120** (FIG. **13**B) configured to drain oil from the SRFF assembly **10**.

(28) In the example embodiment, the latch cage chamber **108** is generally on a volume ‘V’ (FIG. **9**) of the latch pin **100**. The finger **112** of the latch cage **102** includes a shelf region **122**, and a flat **124** defined on the latch pin **100** in the latch cage chamber **108** is configured to engage the shelf region **122** of the latch cage **102**. As shown in FIG. **11**, rotation of the latch pin **100** is stopped by engagement between the flat **124** of the latch pin **100** and the shelf region **122** of the latch cage **102**. Advantageously, in the example implementation, the diameter of the latch biasing member **104** can be increased, thus reducing the spring rate and improving the performance of the latch biasing member **104**.

(29) Described herein are systems and methods for a switching roller finger follower assembly having unique longitudinal inner arm bends and stopping features with a compact latching design. The longitudinal bends are spaced apart and include a reaction pin to limit rotation of the lost motion springs, which are located at ends of the inner arm. Rotation of the inner arm is prevented utilizing material on the outer arm pad or a pin on the outer arm. Moreover, two additional folding operations reduce the width of the inner arm in the valve area. An anti-rotation pin on the inner arm is crimped in the middle to deform the pin in the area where the pin is in contact with the lost motion spring. A shelf on the latch cage is clearance-fit on the latch pin internal chamber, and the shelf region is utilized to stop rotation of the latch pin. The unique geometry of the stamped inner arm advantageously improves stiffness and packaging, avoids contact between the longitudinal bends, and reduces contact stress on the lost motion springs. Further, the latch pin and latch cage

design improves packaging to thereby improve latch compression spring performance and response time of the system.

(30) The foregoing description of the examples has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular example are generally not limited to that particular example, but, where applicable, are interchangeable and can be used in a selected example, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

Claims

1. A switching roller finger follower (SRFF) assembly for valve actuation, the SRFF assembly comprising: an outer arm; an inner arm pivotally coupled to the outer arm via a pivot axle and at least partially disposed within the outer arm, the inner arm being unitarily formed and having a first inner side arm and a second inner side arm, the first inner side arm and the second inner side arm proximate to a first end of the inner arm being bent toward each other such that a first spacing between the first and second inner side arms proximate to the first end is less than a second spacing between the first and second inner side arms proximate to a second end of the inner arm; a spring engagement pin extending through pin apertures formed in the first and second inner side arms proximate to the first end of the inner arm; a pair of lost motion springs supported by the pivot axle; an inner roller rotatably coupled to the inner arm; a pair of outer rollers disposed on respective axles supported by the outer arm; a latch assembly configured to selectively latch the inner arm to the outer arm to prevent relative movement therebetween, wherein the latch assembly comprises a latch pin slidingly received within a latch bore formed in the outer arm, wherein the outer arm includes a hydraulic port fluidly coupled to the latch bore, and wherein said first spacing is defined by respective inner surfaces of the first and second inner side arms being offset from each other by a predetermined distance such that said respective inner surfaces do not contact each other.

2. The SRFF assembly of claim 1, wherein the spring engagement pin defines crimped regions thereon.

3. The SRFF assembly of claim 2, wherein each lost motion spring of the pair of lost motion springs has a first end that engages the respective crimped regions of the spring engagement pin, the crimped regions reducing contact stress on the respective lost motion springs.

4. The SRFF assembly of claim 1, wherein the first inner side arm and the second inner side arm each have a bushing aperture to receive a bushing and an inner roller axle.

5. The SRFF assembly of claim 4, wherein the inner roller is rotatably coupled to the inner arm by the bushing.

6. The SRFF assembly of claim 1, wherein the outer arm comprises a stop thereon configured to stop rotation of the inner arm relative to the outer arm.

7. The SRFF assembly of claim 6, wherein the stop comprises material on the outer arm defining a shoulder to be contacted by the inner arm.

8. The SRFF assembly of claim 6, wherein the stop comprises a stop pin extending from the outer arm.

9. The SRFF assembly of claim 1, wherein the hydraulic port is configured to selectively receive pressurized fluid from a hydraulic lash adjuster.

10. The SRFF assembly of claim 1, wherein the latch assembly further comprises a latch cage operably associated with the latch pin.

11. The SRFF assembly of claim 10, wherein the latch cage is biased away from the latch pin via a spring.

12. The SRFF assembly of claim 10, wherein the latch pin defines a latch cage chamber configured to selectively receive at least a portion of a finger of the latch cage.
13. The SRFF assembly of claim 12, wherein the latch cage finger includes a shelf region and wherein the latch pin comprises a flat, wherein the flat is configured to engage the shelf region of the latch cage and inhibit rotation of the latch pin.
14. A switching roller finger follower (SRFF) assembly for valve actuation, the SRFF assembly comprising: an outer arm; an inner arm pivotally coupled to the outer arm via a pivot axle and at least partially disposed within the outer arm, the inner arm being unitarily formed and having a first inner side arm and a second inner side arm, the first inner side arm and the second inner side arm proximate to a first end of the inner arm being bent toward each other such that a first spacing between the first and second inner side arms proximate to the first end is less than a second spacing between the first and second inner side arms proximate to a second end of the inner arm; a spring engagement pin extending through pin apertures formed in the first and second inner side arms proximate to the first end of the inner arm and defining crimped regions thereon; a pair of lost motion springs supported by the pivot axle and having ends that engage the respective crimped regions of the spring engagement pin, the crimped regions reducing contact stress on the respective lost motion springs; an inner roller coupled to the inner arm by a bushing; and a pair of outer rollers disposed on respective axles supported by the outer arm; a latch assembly configured to selectively latch the inner arm to the outer arm to prevent relative movement therebetween, the latch assembly comprising: a latch pin slidingly received within a latch bore formed in the outer arm, the latch pin defining an internal latch cage chamber and a flat; a latch cage having a finger configured to be at least partially received within the latch cage chamber, the finger including a shelf region, wherein the flat is configured to engage the shelf region of the latch cage and inhibit rotation of the latch pin; and a spring configured to bias the latch pin away from the latch cage, wherein said first spacing is defined by respective inner surfaces of the first and second inner side arms being offset from each other by a predetermined distance such that said respective inner surfaces do not contact each other.
15. A switching roller finger follower (SRFF) assembly for valve actuation, the SRFF assembly comprising: an outer arm; a unitarily formed inner arm including a first inner side arm defining a first roller portion, a first inner roller aperture formed in the first roller portion, a first arm portion, a first pin aperture formed in the first arm portion, and a first pivot axle aperture formed in the first arm portion between the first pin aperture and the first inner roller aperture, and a second inner side arm defining a second roller portion, a second inner roller aperture formed in the second roller portion, a second arm portion, a second pin aperture formed in the second arm portion, and a second pivot axle aperture formed in the second arm portion between the second pin aperture and the second inner roller aperture, wherein a first distance is defined between the first arm portion and the second arm portion, wherein a second distance, greater than the first distance, is defined between the first roller portion and the second roller portion, and wherein the first arm portion does not contact the second arm portion; a pivot axle extending through the first pivot axle aperture and the second pivot axle aperture and coupled to the outer arm to pivotably mount the inner arm to the outer arm; a spring engagement pin extending through the first pin aperture and the second pin aperture; a pair of lost motion springs supported by the pivot axle and engaged between the outer arm and the spring engagement pin; an inner roller rotatably coupled to the first inner roller aperture and the second inner roller aperture; and a pair of outer rollers disposed on respective axles supported by the outer arm.
16. The SRFF assembly of claim 15, further comprising a latch assembly configured to selectively latch the inner arm to the outer arm to prevent relative movement therebetween.
17. The SRFF assembly of claim 16, wherein the latch assembly includes a latch pin slidingly received within a latch bore formed in the outer arm, and wherein the outer arm includes a hydraulic port fluidly coupled to the latch bore.

18. The SRFF assembly of claim 17, wherein the latch assembly further includes a latch cage operably associated with the latch pin, wherein the latch cage is biased away from the latch pin via a biasing spring, and wherein the latch pin defines a latch cage chamber configured to selectively receive at least a portion of a finger of the latch cage.
19. The SRFF assembly of claim 15, wherein the first inner side arm defines a first longitudinal bend between the first roller portion and the first arm portion, and wherein the second inner side arm defines a second longitudinal bend between the second roller portion and the second arm portion.
20. The SRFF assembly of claim 15, wherein the pair of lost motion springs are positioned outboard of the outer arm.